

SOP-Guard

AI-Powered Clinical SOP Assistant with Built-in Safety Verification

What Is This Project?

SOP-Guard is an AI assistant that helps doctors and nurses quickly find answers from hospital Standard Operating Procedures (SOPs). Instead of searching through long PDF documents under time pressure, a clinician simply asks a question in plain language and gets a direct, sourced answer. The key innovation is that the system **checks its own answer** before showing it — verifying that dosages, step sequences, and contraindications are correct according to the official SOP.

What Problem Does It Solve?

Hospitals have hundreds of SOPs that are long, frequently updated, and hard to search. Clinicians — especially during night shifts or emergencies — often need a specific detail (a dosage limit, a required step) but must manually dig through documents to find it. This wastes time and increases the risk of errors. SOP-Guard makes this information instantly accessible and verifiably accurate.

How Does It Work?

- The clinician types a question or **records a voice message** (e.g., "What is the sepsis protocol for patients with renal impairment?"). Audio is transcribed using Whisper and processed like text.
- The system searches the SOP database using AI-powered retrieval (RAG) to find relevant sections.
- A reasoning agent generates an answer grounded in those sections, with citations.
- A **Procedural Faithfulness Verifier** automatically checks the answer for three error types: wrong thresholds, skipped steps, and missed contraindications.
- The answer is shown with a confidence score, source highlights, and reasoning trace. If confidence is low, the system says so instead of guessing.

Technology Used

LLM: Claude / Llama 3.1 (can run locally inside hospital network) | **Voice Input:** OpenAI Whisper (speech-to-text) | **Orchestration:** LangGraph (multi-agent pipeline) | **Retrieval:** Qdrant vector database + BM25 hybrid search | **Backend:** FastAPI | **Frontend:** React (mobile-friendly) | **Evaluation:** RAGAS + custom procedural faithfulness metrics

Voice Input for Hands-Free Use

Doctors and nurses often have their hands occupied — during procedures, while wearing gloves, or at the bedside. SOP-Guard supports **audio input**: clinicians tap a microphone button, speak their question, and the system transcribes it using OpenAI Whisper (fine-tuned for medical terminology) and processes it like any typed query. The transcribed text is shown for confirmation before submitting, reducing the risk of misheard medical terms.

Clinician-Driven SOP Updates (Living Knowledge Base)

SOPs go stale — new research emerges, protocols change, and frontline staff learn things that never make it back into the documents. SOP-Guard includes a **structured update portal** where authorized clinicians (department heads, quality officers) can propose updates to specific SOP sections. Each update is tagged with the date, the clinician's name, and a brief reason for the change. Updates go through an approval workflow before going live. The system maintains a **version history** — so anyone can see what changed, when, and why. The AI knowledge base is automatically re-indexed after each approved update, ensuring answers always reflect the latest protocols.

What Makes It Novel?

Existing clinical QA systems retrieve and generate answers but have no way to verify procedural correctness. SOP-Guard introduces a **Procedural Faithfulness Verifier** — a dedicated component that checks whether the AI's answer follows the exact rules in the SOP (correct numbers, correct order, no missed exceptions). This verify-then-respond pattern is our core research contribution.

Why Is It Practical?

- **No EHR integration needed** — works only with SOP documents, avoiding data privacy barriers.
- **Runs on-premise** — nothing leaves the hospital network.
- **Deployable in weeks** — not years. Start with one department, expand based on feedback.
- **Clinician feedback loop** — doctors can flag incorrect answers, improving the system over time.